

A WILLMORE INEQUALITY FOR IMMERSED CAPILLARY SURFACES OF ANY CO-DIMENSIONS

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ABSTRACT. In this note we prove a Willmore inequality for immersed capillary surfaces of any co-dimensions, which is in fact established in a more general class, namely, the so-called enhanced oriented capillary integral 2-varifolds.

We consider a (possibly unbounded) domain $\Omega \subset \mathbb{R}^n$ ($n \geq 3$) with C^2 boundary $S = \partial\Omega$. Denote by ν^S the inward unit normal of Ω along S . For $X, Y \in T_x S$, the second fundamental form is given by $h_S(X, Y) = -\langle D_X \nu^S, Y \rangle$. Let $\beta \in (0, \pi)$ be a constant which will always be used to prescribe the capillary angle.

Theorem 1. *For any smooth, compact, capillary immersion $F : M \rightarrow \overline{\Omega}$ with boundary in S , the Willmore inequality (15) below holds, and equality happens iff (16) below holds.*

We will prove Theorem 1 in a more general GMT settings in the spirit of Wang-Zhang [7] and myself [8], let us first recall some notations: The unoriented and oriented Grassmannians are denoted as $G(2, n)$ and

$$\tilde{G}(2, n) = \{\xi \in \Lambda_2(\mathbb{R}^n) : \xi \text{ is simple, } |\xi| = 1\}.$$

An oriented 2-varifold is a positive Radon measure V on $\mathbb{R}^n \times \tilde{G}(2, n)$. Let $P(\xi)$ be the orthogonal projection onto the plane represented by ξ , set $P(\xi)^\perp = I - P(\xi)$, and write

$$\mathbf{q}(x, \xi) = (x, P(\xi)), \quad \pi(x, \xi) = x.$$

The associated unoriented varifold of V is denoted as $V_0 = \mathbf{q}_\# V$, its *weight measure* is $\mu_V = \pi_\# V = \|V_0\|$, and its first variation is given by

$$\delta V_0(X) = \int \text{tr}(P(\xi)DX(x))dV(x, \xi), \quad X \in C_c^1(\mathbb{R}^n; \mathbb{R}^n).$$

Following Hutchinson [2] we call the oriented varifold V *integral* if there exist a countably 2-rectifiable set R , a measurable unit orientation ξ_R of its approximate tangent planes, and nonnegative integer-valued multiplicities θ_+, θ_- , with locally integrable positive sum, such that for any $\varphi \in C_c^0(\mathbb{R}^n \times \Lambda_2(\mathbb{R}^n))$,

$$V(\varphi) = \int_R (\theta_+(x)\varphi(x, \xi_R(x)) + \theta_-(x)\varphi(x, -\xi_R(x)))d\mathcal{H}^2(x).$$

Associated to V there is a 2-current $\mathbf{c}(V)$, defined by its action on any 2-form ω having compact support in $\mathbb{R}^n$ as follows:

$$(1) \quad \mathbf{c}(V)(\omega) = \int \langle \xi, \omega(x) \rangle dV(x, \xi),$$

here $\langle \cdot, \cdot \rangle$ denotes the dual paring.

Definition 2 (Oriented capillary bundle [7, 8]). Under the above notations and assumptions, we define the *oriented capillary bundle* as

$$\tilde{G}_{2,\beta}(S) = \left\{ (x, \xi) \in S \times \tilde{G}(2, n) : |P(\xi)(\nu^S(x))| = \sin \beta \right\},$$

endowed with the subspace topology from $S \times \tilde{G}(2, n)$. For any $x \in S$, we also write the fiber as

$$\tilde{G}_{2,\beta}(x) = \left\{ \xi \in \tilde{G}(2, n) : |P(\xi)(\nu^S(x))| = \sin \beta \right\}.$$

For any $(x, \xi) \in \tilde{G}_{2,\beta}(S)$ we define a unit vector field

$$n_\beta(x, \xi) := \frac{P(\xi)(\nu^S(x))}{\sin \beta},$$

which is well-defined thanks to $\sin \beta > 0$, and we call it the *measure-theoretic inward conormal*. We can then define the *measure-theoretic boundary tangent field* as

$$\tau_\beta(x, \xi) := -\iota_{n_\beta(x, \xi)} \xi,$$

where $\iota_w(u \wedge v) = \langle w, u \rangle v - \langle w, v \rangle u$. Clearly, $\tau(x, \xi)$ is unit, $\text{span}\{\tau(x, \xi)\} = \text{im } P(\xi) \cap T_x S$, and $\xi = \tau(x, \xi) \wedge n_\beta(x, \xi)$. We also define the auxiliary field

$$z_\beta(x, \xi) = \begin{cases} \frac{n_\beta(x, \xi) - \sin \beta \nu^S(x)}{\cos \beta}, & \cos \beta \neq 0, \\ 0, & \beta = \pi/2, \end{cases}$$

and note that by construction, $z_\beta(x, \xi)$ is a unit vector in $T_x S$ perpendicular to $\tau(x, \xi)$, when $\cos \beta \neq 0$. For every $\beta \in (0, \pi)$ we can write

$$n_\beta(x, \xi) = \sin \beta \nu^S(x) + \cos \beta z_\beta(x, \xi).$$

We generalize the capillary varifolds in [7] to the following settings.

Definition 3 (Oriented capillary varifolds and enhancement). Let V be an oriented 2-varifold compactly supported on $\bar{\Omega}$, let Γ be a Radon measure with compact support on $\tilde{G}_{2,\beta}(S)$. We call (V, Γ) an *oriented capillary varifold* if there exists a V -measurable field $\mathbf{h} \in L^2(V; \mathbb{R}^n)$, with $P(\xi)(\mathbf{h}(x, \xi)) = 0$ for V -almost everywhere such that the *Gauss-Green*

formula¹ holds:

$$(2) \quad \begin{aligned} \delta V_0(X) &= - \int \langle \mathbf{h}(x, \xi), X(x) \rangle dV(x, \xi) - \int \langle n_\beta(x, \xi), X(x) \rangle d\Gamma(x, \xi) \\ &= - \int \langle H_V(x), X(x) \rangle d\mu_V(x) - \int \langle \mathbf{n}_\Gamma(x), X(x) \rangle d\sigma(x), \quad \forall X \in C_c^1(\mathbb{R}^N; \mathbb{R}^N), \end{aligned}$$

where $H_V := \frac{d\pi_\#(\mathbf{h}V)}{d\mu_V}$, $\sigma := \pi_\#\Gamma$, and

$$(3) \quad \mathbf{n}_\Gamma = \frac{d\pi_\#(n_\beta\Gamma)}{d\sigma} = \sin \beta \nu^S + \cos \beta \frac{d\pi_\#(z_\beta\Gamma)}{d\sigma}.$$

An oriented capillary varifold is additionally called *enhanced*² if V is integral and the current $\mathbf{c}(V) \in \mathbf{I}_2(\mathbb{R}^n)$ given by (1) satisfies for any smooth 1-form α with compact support:

$$(4) \quad \partial \mathbf{c}(V)(\alpha) = \int \alpha(\tau_\beta(x, \xi)) d\Gamma(x, \xi),$$

and there exists an integral (unoriented) 1-varifold Σ such that $\|\Sigma\| = \sigma = \pi_\#\Gamma$, the approximate tangent space $T_x\Sigma = \text{span}\{\tau_\beta(x, \xi)\}$ for Γ -a.e. (x, ξ) , and there exists some $\kappa_\Sigma \in L^1(\sigma; \mathbb{R}^n)$ such that

$$(5) \quad \delta \Sigma(X) = - \int \langle \kappa_\Sigma(x), X(x) \rangle d\sigma(x), \quad \forall X \in C_c^1(\mathbb{R}^n; \mathbb{R}^n).$$

Remark 4. It is direct to check that any smooth, compact capillary immersion $F : M \rightarrow \overline{\Omega}$ with boundary in S induces an enhanced oriented capillary varifold (V, Γ) in the sense of Definition 3. In particular, (4) says that the boundary of the induced current of M , denoted as $F_\# \llbracket M \rrbracket$, agrees with the induced current of ∂M , which is $F_\# \llbracket \partial M \rrbracket$. While (5) is just the first variation of ∂M , with κ_Σ the boundary mean curvature vector.

Inspired by [6], we define the capillary Willmore energy for (V, Γ) as follows.

Definition 5. For an enhanced oriented capillary varifold (V, Γ) in the sense of Definition 3, its Willmore energy is defined as

$$(6) \quad \mathcal{W}_0(V, \Gamma) := \frac{1}{4} \int |H_V|^2 d\mu_V + \int \langle \kappa_\Sigma, \mathbf{n}_\Gamma \rangle d\sigma,$$

for any $\ell \in \mathbb{Z}$ fixed we also put

$$\mathcal{W}_{\beta, \ell} := \mathcal{W}_0 - 2\pi\ell \cos \beta.$$

¹This is similar to the formulation of [7, Definition 1.2], but with the stronger requirement that the test fields are arbitrary ambient fields, rather than just being tangential along S .

²I use the notion "enhanced" somewhat inspired by Ilmanen's notion of the so-called *enhanced motion* in [3].

Proposition 6. *The field H_V is the generalized mean curvature of V_0 , and both energies in (6) are finite. Write $\kappa_\Sigma^S = P_{TS}(\kappa_\Sigma)$, one has*

$$(7) \quad \mathcal{W}_{\beta,\ell} = \frac{1}{4} \int |H_V|^2 d\mu_V + \sin \beta \int h_S(\tau, \tau) d\sigma + \cos \beta \int \langle \kappa_\Sigma^S, d(\pi_\#(z_\beta \Gamma)) \rangle - 2\pi\ell \cos \beta.$$

Here τ can be taken as any one of the two choices of unit vectors spanning $T_x \Sigma$.

Proof. By the assumptions that $\mathbf{h} \in L^2(V; \mathbb{R}^n)$ and $\kappa_\Sigma \in L^1(\sigma; \mathbb{R}^n)$ we infer easily that $H_V \in L^2(\mu_V)$, and that both energies in (6) are finite. Since σ is 1-rectifiable and μ_V is 2-rectifiable, we know $\sigma \perp \mu_V$. It follows from (2) that H_V is the generalized mean curvature of V_0 . Moreover, by Brakke's perpendicular theorem ([1, Sect. 5.8]) we see that H_V is perpendicular to the approximate tangent space of μ_V at μ_V -a.e.

Now we show that $\mathbf{n}_\Gamma \perp T_x \Sigma$ and $|\mathbf{n}_\Gamma| \leq 1$. To see this, note that by disintegration we can write $\Gamma = \sigma \otimes \Gamma^x$, where Γ^x is a Radon probability measure on $\tilde{G}_{2,\beta}(x)$ for σ -a.e. x . Back to the definition of $\mathbf{n}_\Gamma$, we can rewrite it as

$$\mathbf{n}_\Gamma(x) = \int n_\beta(x, \xi) d\Gamma^x(\xi),$$

which implies directly that $|\mathbf{n}_\Gamma| \leq 1$. Moreover, by the assumption that $T_x \Sigma = \text{span}\{\tau_\beta(x, \xi)\}$ for Γ -a.e., (x, ξ) , taking also the fact that $\langle n_\beta(x, \xi), \tau_\beta(x, \xi) \rangle = 0$, we thus infer

$$\mathbf{n}_\Gamma \perp T_x \Sigma.$$

Finally, for any $\phi \in C_c^1(\mathbb{R}^n)$, testing (5) with a compactly supported C^1 -extension of $\phi \nu^S$ yields

$$- \int \phi h_S(\tau, \tau) d\sigma = \delta \Sigma(\phi \nu^S) = - \int \phi \langle \kappa_\Sigma, \nu^S \rangle d\sigma,$$

where we have used the fact that

$$\text{div}_{T_x \Sigma}(\phi \nu^S) = \langle D_\tau(\phi \nu^S), \tau \rangle = D_\tau \phi \langle \nu^S, \tau \rangle + \phi \langle D_\tau \nu^S, \tau \rangle = -\phi h_S(\tau, \tau).$$

It is then standard to deduce that $\langle \kappa_\Sigma, \nu^S \rangle = h_S(\tau, \tau)$ for σ -a.e. Combining with (3) we deduce (7), and the proof is completed. $\square$

We prove the Willmore inequality in the spirit of Simon [5], let us first introduce some notations: for $a \in S$ we denote the inversion at a as

$$Y_a(x) = \frac{x - a}{|x - a|^2} \quad (x \neq a).$$

For $\varepsilon \ll 1$, we also denote the regularization

$$Y_{a,\varepsilon}(x) := \frac{x - a}{\max\{|x - a|, \varepsilon\}^2}.$$

Whenever the finite limit exists, set

$$(8) \quad L_a = \lim_{\varepsilon \searrow 0} \int \langle Y_{a,\varepsilon}, d(\pi_\#(z_\beta \Gamma)) \rangle,$$

and

$$\mathcal{D}_a = \int \langle \kappa_\Sigma, d(\pi_\#(z_\beta \Gamma)) \rangle + 2L_a.$$

We denote the density of μ_V as

$$\Theta^2(V, a) := \lim_{r \searrow 0} \frac{\mu_V(B_r(a))}{\pi r^2}$$

whenever it exists.

We can now prove the following Simon-type monotonicity formula:

Lemma 7. *Let (V, Γ) be an enhanced oriented capillary varifold in the sense of Definition 3. Assume that μ_V and σ are compactly supported. For any $a \in S$ and $t > 0$, define*

$$\begin{aligned} \mathcal{M}_a(t) &= \frac{\mu_V(B_t(a))}{t^2} + \frac{1}{2t^2} \int_{B_t(a)} \langle H_V, x - a \rangle d\mu_V \\ &\quad + \frac{1}{16} \int_{B_t(a)} |H_V|^2 d\mu_V + \frac{1}{2} \int \langle Y_{a,t}, \mathbf{n}_\Gamma \rangle d\sigma. \end{aligned}$$

Then, for every $0 < r < R$,

$$(9) \quad \mathcal{M}_a(R) - \mathcal{M}_a(r) = \int_{B_R(a) \setminus B_r(a)} \left| \frac{1}{4} H_V + P^\perp Y_a \right|^2 d\mu_V \geq 0.$$

Here $P = P(x)$ is the approximate tangent projection of V_0 .

Proof. Fix $a \in S$ and first choose $0 < r < R$ such that $\mu_V(\partial B_r(a)) = \mu_V(\partial B_R(a)) = 0$. As in [5] we consider the test vector field

$$X(x) = \left(\frac{1}{\max\{|x - a|, r\}^2} - \frac{1}{R^2} \right)_+ (x - a) = Y_{a,r}(x) - Y_{a,R}(x),$$

which is Lipschitz and supported in $\overline{B_R(a)}$. By a standard approximation argument, we can test (2) with such X . A direct computation then yields

$$\operatorname{div}_P X(x) = \begin{cases} 2(r^{-2} - R^{-2}), & |x - a| < r, \\ 2|P^\perp(Y_a(x))|^2 - 2R^{-2}, & r < |x - a| < R, \\ 0, & |x - a| > R. \end{cases}$$

Thus (2) gives

$$\begin{aligned} (10) \quad & \frac{2\mu_V(B_r(a))}{r^2} - \frac{2\mu_V(B_R(a))}{R^2} + 2 \int_{B_R(a) \setminus B_r(a)} |P^\perp(Y_a)|^2 d\mu_V \\ &= -\frac{1}{r^2} \int_{B_r(a)} \langle H_V, x - a \rangle d\mu_V + \frac{1}{R^2} \int_{B_R(a)} \langle H_V, x - a \rangle d\mu_V \\ &\quad - \int_{B_R(a) \setminus B_r(a)} \langle H_V, Y_a \rangle d\mu_V - \int \langle Y_{a,r} - Y_{a,R}, \mathbf{n}_\Gamma \rangle d\sigma. \end{aligned}$$

Since $P(H_V) = 0$ μ_V -almost everywhere, at any such point we can rewrite

$$|P^\perp(Y_a)|^2 + \frac{1}{2}\langle H_V, Y_a \rangle = \left| P^\perp(Y_a) + \frac{1}{4}H_V \right|^2 - \frac{1}{16}|H_V|^2.$$

Substituting this into (10) proves (9). □

Lemma 8. *Under the hypotheses of Lemma 7, the following assertions hold. For σ -almost every $a \in S$, the finite limit L_a defined in (8) exists and equals*

$$L_a = \lim_{\varepsilon \searrow 0} \int_{\{|x-a|>\varepsilon\}} \langle Y_a(x), z_\beta(x, \xi) \rangle d\Gamma(x, \xi).$$

At any point a as above, the density $\Theta^2(V, a)$ exists and is finite. Moreover,

$$(11) \quad \begin{aligned} \mathcal{W}_{\beta, \ell}(V, \Gamma) &= 4\pi\Theta^2(V, a) - 2\pi\ell \cos \beta + \frac{1}{4} \int |H_V + 4P^\perp(Y_a)|^2 d\mu_V \\ &\quad + \cos \beta \mathcal{D}_a + \sin \beta \int [h_S(\tau, \tau) + 2\langle Y_a, \nu^S \rangle] d\sigma. \end{aligned}$$

Sketch of proof. Following [5], we take $r \searrow 0$ and $R \rightarrow \infty$ in Lemma 7. The terms involving μ_V can be handled exactly as in [5], therefore it suffices to consider the boundary limit. We notice that since σ has compact support, by the dominated convergence and (3),

$$\lim_{r \searrow 0} \int \langle Y_{a,r}, \mathbf{n}_\Gamma \rangle d\sigma = \sin \beta \int \langle Y_a, \nu^S \rangle d\sigma + \cos \beta L_a.$$

On the other hand, for any sufficiently large R ,

$$\int \langle Y_{a,R}, \mathbf{n}_\Gamma \rangle d\sigma = R^{-2} \int \langle x - a, \mathbf{n}_\Gamma \rangle d\sigma \rightarrow 0.$$

We thus conclude (11). □

Lemma 9. *Under the assumptions of Lemma 8, there is a set $\mathcal{A} \subset S$ of full $\|\partial \mathbf{c}(V)\|$ -measure such that for any $a \in \mathcal{A}$, L_a is finite and*

$$(12) \quad \Theta^2(V, a) \geq \frac{k(a)}{2},$$

where $k(a) \in \mathbb{N}$ is the multiplicity of the integral 1-current $\partial \mathbf{c}(V)$. Further, all conclusions of Lemma 8 hold on $\mathcal{A}$.

Proof. By definitions we clearly have $\|\mathbf{c}(V)\| \leq \mu_V$ and $\|\partial \mathbf{c}(V)\| \leq \sigma$. We choose a set $\mathcal{A}$ of full $\|\partial \mathbf{c}(V)\|$ -measure on which Lemma 8 holds, and the boundary current $\partial \mathbf{c}(V)$ and its mass measure $\|\partial \mathbf{c}(V)\|$ have their multiplicity- $k(a)$ approximate tangent space.

For any fixed $a \in \mathcal{A}$, we let L be the oriented boundary tangent line. Let $D_{a,r}(x) = (x - a)/r$ denote the homothety map, let $r_j \searrow 0$ be a sequence of numbers then set

$\mathbf{c}_j := (D_{a,r_j})_{\#} \mathbf{c}(V)$ and $\mu_{V_j} = r_j^{-2} (D_{a,r_j})_{\#} \mu_V$. For every fixed $R > 0$, finite density and convergence of the rescaled boundary mass measures give

$$\|\mathbf{c}_j\|(B_R) \leq \mu_{V_j}(B_R) = \frac{\mu_V(B_{Rr_j}(a))}{r_j^2} = \pi R^2 \Theta^2(V, a) + o(1),$$

and also

$$\|\partial \mathbf{c}_j\|(B_R) = \frac{\|\partial \mathbf{c}(V)\|(B_{Rr_j}(a))}{r_j} = 2k(a)R + o(1),$$

where we used $\partial \mathbf{c}_j = (D_{a,r_j})_{\#} \partial \mathbf{c}(V)$. Thus the currents $\mathbf{c}_j$ and their boundaries have uniformly bounded mass on every fixed B_R . By the integral-current compactness theorem [4, Chapter 6] we deduce that, after passing to a subsequence (still indexed by $j \in \mathbb{N}$), $\mathbf{c}_j \rightarrow \mathbf{c}_{\infty}$ as currents and $\mu_{V_j} \xrightarrow{*} \mu_{\infty}$ as measures, with $\partial \mathbf{c}_{\infty} = k(a)[L]$, $\|\mathbf{c}_{\infty}\| \leq \mu_{\infty}$, and

$$(13) \quad \mu_{\infty}(B_R) \leq \pi \Theta^2(V, a) R^2 \quad \forall R > 0.$$

By the slicing theorem [4, Section 28], for almost every $\rho > 0$, the integral slice $S_{\rho} = \langle \mathbf{c}_{\infty}, |x|, \rho \rangle$ is tangent to ∂B_{ρ} and satisfies

$$(14) \quad \partial S_{\rho} = -\partial((\partial \mathbf{c}_{\infty}) \lrcorner B_{\rho}) = -k(a) \partial([L] \lrcorner B_{\rho}) = -k(a)(\delta_{\rho e} - \delta_{-\rho e}),$$

where e is a unit vector spanning L and agrees with the orientation of L . For $0 < \varepsilon < 1$, define on ∂B_{ρ} the function

$$f_{\varepsilon}(y) = \rho \arccos \left((1 - \varepsilon) \frac{\langle y, e \rangle}{\rho} \right).$$

A direct computation gives $|\nabla^{\partial B_{\rho}} f_{\varepsilon}| \leq 1$. Since the integral slice S_{ρ} is tangent to ∂B_{ρ} ,

$$|S_{\rho}(\mathrm{d}f_{\varepsilon})| = \left| \int \left\langle \nabla^{\partial B_{\rho}} f_{\varepsilon}, \vec{S}_{\rho} \right\rangle \mathrm{d}\|S_{\rho}\| \right| \leq \int 1 \mathrm{d}\|S_{\rho}\| = \mathbf{M}(S_{\rho}),$$

where we have used the polar decomposition of S_{ρ} . It follows that

$$\mathbf{M}(S_{\rho}) \geq |S_{\rho}(\mathrm{d}f_{\varepsilon})| = |\partial S_{\rho}(f_{\varepsilon})| \stackrel{(14)}{=} k\rho[\pi - 2\arccos(1 - \varepsilon)].$$

Letting $\varepsilon \searrow 0$, we get $\mathbf{M}(S_{\rho}) \geq k\pi\rho$. By the coarea formula [4, (28.10)], we find

$$\pi \Theta^2(V, a) R^2 \stackrel{(13)}{\geq} \mu_{\infty}(B_R) \geq \|\mathbf{c}_{\infty}\|(B_R) \geq \int_0^R \mathbf{M}(S_{\rho}) \mathrm{d}\rho \geq \frac{k\pi R^2}{2}.$$

The assertion (12) then follows. $\square$

Theorem 10 (Willmore inequality for enhanced oriented capillary varifolds). *Fix $\ell \in \mathbb{Z}$. Let (V, Γ) be an enhanced oriented capillary varifold in the sense of Definition 3. Let $\mathcal{A}$ be the set resulting from Lemma 9. For any $a \in \mathcal{A}$, one has*

$$(15) \quad \mathcal{W}_{\beta, \ell}(V, \Gamma) \geq 2\pi(k(a) - \ell \cos \beta) + \cos \beta \mathcal{D}_a + \sin \beta \int [h_S(\tau, \tau) + 2\langle Y_a, \nu^S \rangle] \mathrm{d}\sigma,$$

where k_a is the multiplicity as in Lemma 9. For each such a , equality holds if and only if

$$(16) \quad \Theta^2(V, a) = \frac{k(a)}{2}, \quad H_V(x) = -4P(x)^\perp(Y_a(x)) \quad \mu_V\text{-a.e.}$$

Proof. Fix $a \in \mathcal{A}$. Subtracting the right-hand side of (15) from (11) gives exactly

$$4\pi \left(\Theta^2(V, a) - \frac{k(a)}{2} \right) + \frac{1}{4} \int |H_V + 4P^\perp(Y_a)|^2 d\mu_V,$$

where the first term is nonnegative by Lemma 9, and the second is clearly nonnegative. The sum of these two terms vanishes precisely when (16) holds. Combining, (15) and its equality case is thus proved. $\square$

Proof of Theorem 1. The result follows from Theorem 10, thanks to Remark 4. $\square$

Remark 11. Let $F : M \rightarrow \overline{\Omega}$ be a smooth capillary immersion in the setting of [6], where $\Omega = \mathbb{R}_+^3$ or $\mathbb{B}^3$. Taking ℓ as the quantity $\text{ind}(F(\partial M))$ in [6], one can then check that the $\mathcal{W}_{\beta, \ell}$ agrees with the Willmore functional (say $\mathcal{W}_{\text{WXZ}}$) in [6], and the boundary integral $\sin \beta \int [h_S(\tau, \tau) + 2\langle Y_a, \nu^S \rangle] d\sigma$ vanishes. Therefore, if

$$\cos \beta [\mathcal{D}_a - 2\pi(\ell - 1)] \geq 0 \quad \text{for some } a \in \mathcal{A},$$

then (15) yields the same sharp lower bound as in [6]:

$$\mathcal{W}_{\text{WXZ}}(F) \geq 2\pi(1 - \cos \beta).$$

Within the smooth class in [6], equality holds precisely for spherical caps in the half-space, and spherical caps or flat capillary disks in the ball; see [6, Theorem 1.3 and Corollary 3.5].

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REFERENCES

- [1] Kenneth A. Brakke, *The motion of a surface by its mean curvature.*, Math. Notes (Princeton), Princeton University Press, Princeton, NJ, 1978. MR 485012
- [2] John E. Hutchinson, *Second fundamental form for varifolds and the existence of surfaces minimising curvature*, Indiana Univ. Math. J. **35** (1986), no. 1, 45–71. MR 825628
- [3] Tom Ilmanen, *Elliptic regularization and partial regularity for motion by mean curvature*, Mem. Amer. Math. Soc. **108** (1994), no. 520, x+90. MR 1196160
- [4] Leon Simon, *Lectures on geometric measure theory*, Proceedings of the Centre for Mathematical Analysis, Australian National University, vol. 3, Australian National University, Centre for Mathematical Analysis, Canberra, 1983. MR 756417
- [5] ———, *Existence of surfaces minimizing the Willmore functional*, Comm. Anal. Geom. **1** (1993), no. 2, 281–326. MR 1243525
- [6] Guofang Wang, Chao Xia, and Xuwen Zhang, *Monotonicity formulas for capillary surfaces*, J. Math. Pures Appl. (9) **204** (2025), Paper No. 103802, 30. MR 4967061
- [7] Guofang Wang and Xuwen Zhang, *Varifolds with capillary boundary*, 2025, arXiv:2503.19052.
- [8] Xuwen Zhang, *Compactness of capillary hypersurfaces with mean curvature prescribed by ambient functions*, Math. Z. **313** (2026), no. 4, Paper No. 74, 23. MR 5101659

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